

# Xianwen Jin

## Software & Data Engineer

📍 Padua, Italy | 📞 +39 388 4953659

✉️ [jinxw@live.it](mailto:jinxw@live.it) | [in LinkedIn](#)

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## SUMMARY

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Software & Data Engineer with 5+ years of experience building high-throughput backend systems, distributed data pipelines, and real-time streaming architectures in enterprise and banking environments. Strong hands-on experience with Apache Flink, Apache Kafka, Change Data Capture (CDC), SQL databases, Kubernetes, and Java/Spring, with a proven track record of improving system scalability, resilience, and performance. Experienced in designing production data pipelines, modernizing legacy data architectures, and troubleshooting distributed systems. Additional expertise in Python, Machine Learning, and LLM Engineering.

## TECHNICAL SKILLS

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**Data Engineering & Streaming:** Apache Flink, Apache Kafka, IBM CDC, Hadoop, Hive, HBase, Real-Time Data Processing, Data Pipelines, Distributed Systems

**Backend Engineering:** Java, Spring, Spring Boot, Quarkus, Microservices, REST APIs, RabbitMQ, JPA, Hibernate

**Databases:** SQL, PostgreSQL, MySQL, Oracle DB, Redis

**Cloud & Infrastructure:** Kubernetes, Docker, Horizontal Pod Autoscaling (HPA), Git, GitLab, Jenkins

**Programming:** Java, Python, Go (Golang), C/C++, SQL, JavaScript

**AI & Machine Learning:** PyTorch, LLM Engineering, RAG, QLoRA Fine-Tuning, AI Agents, NLP

## PROFESSIONAL EXPERIENCE

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### Intesa Sanpaolo

*Software Engineer*

**Turin, Italy**

*Nov 2023 – Present*

- Own backend development and code design on the critical path of the bank's secure authentication lifecycle, supporting highly resilient services used by millions of active users.
- **Streaming Pipeline:** Reengineered a core customer data delivery pipeline from a hybrid legacy setup into a fully decoupled and fault-tolerant Apache Flink streaming architecture, designed to remain resilient under massive concurrent traffic spikes.
- **Authentication Data Service:** Performed deep-dive performance profiling using Java Flight Recorder (JFR) and VisualVM to identify and eliminate microservice bottlenecks in memory allocation and database connection pools.
- Implemented **Horizontal Pod Autoscaling (HPA)** to dynamically scale Kubernetes workloads, improving service scalability and baseline Transactions Per Second (TPS).
- Worked with **Apache Kafka** for high-throughput data streaming and integration between distributed services.
- *Tech Stack:* Java, Spring Boot, Apache Flink, Apache Kafka, Kubernetes, HPA, JFR, VisualVM, Ververica, OracleSQL.

### Huawei

*Software Engineer*

**Dongguan, China**

*Mar 2023 – Aug 2023*

- Developed and maintained core backend systems for **CodeArts Pipeline**, a visual platform for automated software delivery and task scheduling.

- Engineered a background tracking and analytics engine to collect, process, and visualize real-time build, cloud testing, and deployment pipeline statistics across the software development lifecycle.
- Worked with asynchronous messaging and distributed backend components using **RabbitMQ** and Redis.
- *Tech Stack*: Spring Boot, RabbitMQ, Redis, MyBatis, PostgreSQL, MySQL.

## Technology Reply

*Technical Consultant (Data Engineer)*

**Padua, Italy**

*Jan 2021 – Dec 2022*

- **Legacy Data Offloading Pipeline**: Designed and implemented a real-time mainframe data migration pipeline using **Change Data Capture (CDC)**, **Apache Flink**, and **Apache Kafka** for a major Italian banking group supporting PSD2 Open Banking requirements.
- Designed a distributed streaming workflow for continuously transferring and transforming data from legacy mainframe systems into modern data infrastructure.
- Reduced historical database footprints and accelerated relational database query performance by approximately **30%** compared with the legacy DB2 architecture.
- Provided end-to-end technical analysis, application maintenance, production troubleshooting, and direct tier-3 vendor issue resolution.
- *Tech Stack*: Apache Flink, Apache Kafka, IBM CDC, Oracle SQL, Ververica, GitLab, Jenkins.

## CERTIFICATIONS & CONTINUOUS LEARNING

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**AI & Machine Learning**: AI Engineer Core Track (LLM Engineering, RAG, QLoRA, Agents) | Deep Learning: Advanced Natural Language Processing and RNNs | PyTorch for Deep Learning Bootcamp

**Software Engineering**: Go: The Complete Developer's Guide | Working with Concurrency in Go | Starting with Quarkus

## EDUCATION

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**Università degli Studi di Padova**

*Master's Degree in Computer Engineering*

**Padua, Italy**

*Oct 2018 – Dec 2020*

- **Core Focus**: Machine Learning, Big Data Infrastructure, Computer Vision, Algorithm Efficiency.
- **Thesis**: *k-Center Clustering under Doubling Dimension* (Optimizing MapReduce space/time complexities).
- **Prior Education**: Bachelor's Degree in Electronic Engineering (2014 – 2017).

## ACADEMIC & INDEPENDENT PROJECTS

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**MapReduce k-Center Clustering Solver**

*Lead Developer*

**University of Padua**

*Java & Hadoop*

- Implemented a highly optimized distributed clustering solver using the coreset approach and Farthest-First Traversal algorithm.
- Applied the Doubling Dimension concept to reduce computational complexity in distributed MapReduce processing.

**Traveling Salesman Problem (TSP) Optimization Engine**

*Core Programmer*

**University of Padua**

*C & CPLEX*

- Designed an operational research solver using IBM ILOG CPLEX Optimization Studio.
- Formulated Linear Integer Programming models and implemented metaheuristics including 2-opt, Local Branching, and Variable Neighborhood Search.